

```

print(Weekly.describe(include='all'))
print('\nDirection counts:\n', Weekly['Direction'].value_counts())
print('\nCorrelation matrix (numeric columns):\n',
Weekly.select_dtypes(np.number).corr())

fig, axes = plt.subplots(1, 3, figsize=(15,4))
Weekly.groupby('Year')['Volume'].mean().plot(ax=axes[0], title='Avg Volume by
Year')
axes[0].set_xlabel('Year'); axes[0].set_ylabel('Avg Volume')

axes[1].scatter(Weekly['Lag2'], Weekly['Today'], s=10)
axes[1].set_title('Today vs Lag2'); axes[1].set_xlabel('Lag2');
axes[1].set_ylabel('Today')

Weekly['Today'].hist(bins=30, ax=axes[2])
axes[2].set_title('Histogram of Today'); axes[2].set_xlabel('Today return');
axes[2].set_ylabel('count')
plt.tight_layout(); plt.show()

```

	Year	Lag1	Lag2	Lag3	Lag4 \
count	1089.000000	1089.000000	1089.000000	1089.000000	1089.000000
unique	NaN	NaN	NaN	NaN	NaN
top	NaN	NaN	NaN	NaN	NaN
freq	NaN	NaN	NaN	NaN	NaN
mean	2000.048669	0.150585	0.151079	0.147205	0.145818
std	6.033182	2.357013	2.357254	2.360502	2.360279
min	1990.000000	-18.195000	-18.195000	-18.195000	-18.195000
25%	1995.000000	-1.154000	-1.154000	-1.158000	-1.158000
50%	2000.000000	0.241000	0.241000	0.241000	0.238000
75%	2005.000000	1.405000	1.409000	1.409000	1.409000
max	2010.000000	12.026000	12.026000	12.026000	12.026000

	Lag5	Volume	Today	Direction
count	1089.000000	1089.000000	1089.000000	1089
unique	NaN	NaN	NaN	2
top	NaN	NaN	NaN	Up
freq	NaN	NaN	NaN	605
mean	0.139893	1.574618	0.149899	NaN
std	2.361285	1.686636	2.356927	NaN
min	-18.195000	0.087465	-18.195000	NaN
25%	-1.166000	0.332022	-1.154000	NaN
50%	0.234000	1.002680	0.241000	NaN
75%	1.405000	2.053727	1.405000	NaN
max	12.026000	9.328214	12.026000	NaN

```

Direction counts:
Direction
Up      605
Down    484
Name: count, dtype: int64

```

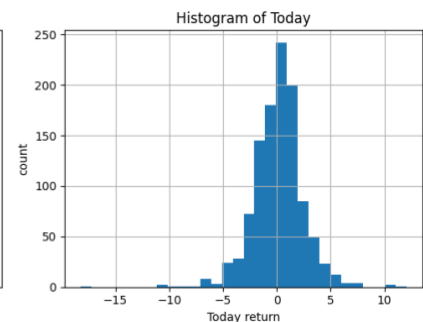
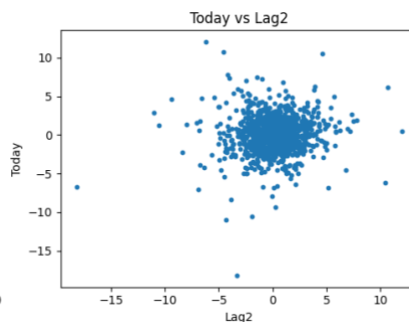
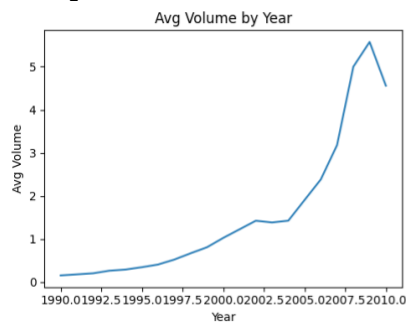
```

Correlation matrix (numeric columns):
      Year      Lag1      Lag2      Lag3      Lag4      Lag5      Volume
\

```

Year	1.000000	-0.032289	-0.033390	-0.030006	-0.031128	-0.030519	0.841942
Lag1	-0.032289	1.000000	-0.074853	0.058636	-0.071274	-0.008183	-0.064951
Lag2	-0.033390	-0.074853	1.000000	-0.075721	0.058382	-0.072499	-0.085513
Lag3	-0.030006	0.058636	-0.075721	1.000000	-0.075396	0.060657	-0.069288
Lag4	-0.031128	-0.071274	0.058382	-0.075396	1.000000	-0.075675	-0.061075
Lag5	-0.030519	-0.008183	-0.072499	0.060657	-0.075675	1.000000	-0.058517
Volume	0.841942	-0.064951	-0.085513	-0.069288	-0.061075	-0.058517	1.000000
Today	-0.032460	-0.075032	0.059167	-0.071244	-0.007826	0.011013	-0.033078

	Today
Year	-0.032460
Lag1	-0.075032
Lag2	0.059167
Lag3	-0.071244
Lag4	-0.007826
Lag5	0.011013
Volume	-0.033078
Today	1.000000



```
Weekly['Direction_bin'] = (Weekly['Direction'] == 'Up').astype(int)
```

```
logit_full = smf.logit("Direction_bin ~ Lag1 + Lag2 + Lag3 + Lag4 + Lag5 +
Volume", data=Weekly).fit()
print(logit_full.summary())
```

Optimization terminated successfully.

Current function value: 0.682441

Iterations 4

Logit Regression Results

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=
Dep. Variable:          Direction_bin   No. Observations:      108
9
```

```

Model:                Logit    Df Residuals:                108
2
Method:                MLE     Df Model:
6
Date:                  Sun, 09 Nov 2025    Pseudo R-squ.:                0.00658
0
Time:                  19:10:25    Log-Likelihood:                -743.1
8
converged:              True     LL-Null:                -748.1
0
Covariance Type:        nonrobust    LLR p-value:                0.131
3

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	coef	std err	z	P> z	[0.025	0.97
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5]

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Intercept	0.2669	0.086	3.106	0.002	0.098	0.43
Lag1	-0.0413	0.026	-1.563	0.118	-0.093	0.01
Lag2	0.0584	0.027	2.175	0.030	0.006	0.11
Lag3	-0.0161	0.027	-0.602	0.547	-0.068	0.03
Lag4	-0.0278	0.026	-1.050	0.294	-0.080	0.02
Lag5	-0.0145	0.026	-0.549	0.583	-0.066	0.03
Volume	-0.0227	0.037	-0.616	0.538	-0.095	0.05

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```

proba_full = logit_full.predict(Weekly)
pred_full = (proba_full >= 0.5).astype(int)

cm_full = confusion_matrix(Weekly['Direction_bin'], pred_full)
acc_full = accuracy_score(Weekly['Direction_bin'], pred_full)

print("Confusion Matrix (Full):\n", cm_full)
print("Accuracy (Full):", acc_full)
print(classification_report(Weekly['Direction_bin'], pred_full,
target_names=['Down', 'Up']))

```

```

Confusion Matrix (Full):
[[ 54 430]
 [ 48 557]]
Accuracy (Full): 0.5610651974288338

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	precision	recall	f1-score	support
Down	0.53	0.11	0.18	484

Up	0.56	0.92	0.70	605
accuracy			0.56	1089
macro avg	0.55	0.52	0.44	1089
weighted avg	0.55	0.56	0.47	1089

```
train = Weekly['Year'] <= 2008
test = Weekly['Year'] >= 2009
```

```
logit_lag2 = smf.logit("Direction_bin ~ Lag2", data=Weekly[train]).fit()
print(logit_lag2.summary())
```

```
proba_te = logit_lag2.predict(Weekly[test])
pred_te = (proba_te >= 0.5).astype(int)
cm_logit = confusion_matrix(Weekly.loc[test, 'Direction_bin'], pred_te)
acc_logit = accuracy_score(Weekly.loc[test, 'Direction_bin'], pred_te)
```

```
print("Logit(Lag2) Test Confusion Matrix:\n", cm_logit)
print("Logit(Lag2) Test Accuracy:", acc_logit)
```

```
Optimization terminated successfully.
Current function value: 0.685555
Iterations 4
```

Logit Regression Results

```
=====
=
Dep. Variable:          Direction_bin   No. Observations:          98
5
Model:                  Logit          Df Residuals:          98
3
Method:                 MLE           Df Model:
1
Date:                  Sun, 09 Nov 2025   Pseudo R-squ.:          0.00307
6
Time:                  19:10:25          Log-Likelihood:          -675.2
7
converged:              True            LL-Null:          -677.3
5
Covariance Type:        nonrobust        LLR p-value:          0.0412
3
=====
```

```
=====
=
coef      std err          z      P>|z|      [0.025      0.97
5]
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-
Intercept    0.2033    0.064     3.162    0.002    0.077    0.32
9
Lag2         0.0581    0.029     2.024    0.043    0.002    0.11
4
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```
Logit(Lag2) Test Confusion Matrix:  
[[ 9 34]  
 [ 5 56]]  
Logit(Lag2) Test Accuracy: 0.625
```